Problem Statement 4 - AI Powered Estimation Portal/Chatbot (Pick Any one of A, B, C or D)

a) AI-Powered Collateral Valuation & Resale Liquidity Engine

Context: Why This Is a Real Problem

For NBFCs and secured lenders, property-backed lending hinges on two fundamental questions:

- **What is the asset worth today?**
- **How easily can it be liquidated if required?**

Most current valuation approaches rely on:

- Manual site inspections
- Broker inputs that are subjective and inconsistent
- Circle rates that often lag actual market conditions

This leads to:

- Large valuation variance
- Mispriced risk and conservative lending
- Slower credit decisioning and turnaround times

Core Problem Statement

Design a **data-led valuation and liquidity engine** that can consistently estimate both **asset value and resale risk** for property-backed lending.

Target Outputs

The system should produce:

- **Estimated Market Value** (₹ range)
- **Distress Sale Value** (₹ range)
- **Resale Potential Index** (0–100)
- **Estimated Time to Liquidate** (days range)
- **Confidence Score** (0–1)
- **Key Risk Flags and Value Drivers**

What You Are Actually Solving

This is **not just a pricing model**.

The objective is to build a **market-aware collateral intelligence layer** that combines:

- **Intrinsic value** (what the asset should sell for)
- **Liquidity** (how fast and how reliably it can be sold)

In practice, lenders care as much about *exit certainty* as valuation.

Feature Engineering Framework

1. **Location Intelligence (Primary Driver)**

REACH
3ROADS
Zudio
REACH
LOOKS

Core Inputs

- Detailed Location: Address / Lat-long
- Photo of exterior and interior (optional, but good to have)

Derived Signals

- **Circle Rate Benchmark**
  Acts as the statutory floor value.
- **Market Activity Proxies**
  Nearby listings, broker density, transaction indicators (where available).
- **Infrastructure Proximity Index**
  Distance to metro/rail, highways, commercial hubs, schools, hospitals.
  Strong driver of both price and resale velocity.
- **Neighbourhood Quality Score**
  Residential vs mixed-use, planned vs unplanned zones.

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1. **Property Characteristics**

Core Inputs

- Property type: Residential / Commercial / Industrial
- Sub-type: Apartment, villa, plot, shop, warehouse
- Size: Carpet / built-up area, land parcel size
- Vintage:
    - New (<5 years)
    - Mid-age (5–15 years)
    - Old (>15 years)
- Floor level and accessibility (lifts, ground floor access)

These factors directly influence:

- Depreciation
- Buyer demand
- Usable life

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3. Legal & Ownership Attributes

Inputs

- Freehold vs leasehold
- Clear title vs potential legal complexity (proxy indicators)

Impact

- Directly affects resale liquidity
- Drives conservative LTV limits in underwriting

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4. Income & Usage Signals

Inputs

- Occupancy status: self-occupied / rented / vacant
- Rental yield proxy (if applicable)

Insight Properties with healthy rental yields tend to attract investors, improving resale certainty.

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5. Market Dynamics Layer

Derived Features

- Supply–demand balance (listing density vs absorption indicators)
- Local price momentum (optional)
- Asset fungibility
    - Standard configurations (e.g., 2BHK apartments) → higher liquidity
    - Custom or niche assets → lower liquidity

Valuation and Liquidity Modeling

1. Market Value Estimation

Market Value is a function of:

- Circle rate benchmark
- Location premium
- Property type and size
- Age and depreciation
- Infrastructure score
- Rental yield (if applicable)

Output: Value range, not a point estimate.

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2. Distress Sale Value

Distress Value = Market Value × Liquidity Discount

Liquidity discounts depend on:

- Asset type
- Location demand
- Legal clarity

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3. Resale Potential Index (0–100)

A consolidated liquidity signal derived from:

Factor	Impact
Prime location	↑
Standard configuration	↑
High demand micro-market	↑

Older construction	↓
Legal complexity	↓
Niche asset profile	↓

Interpretation

- 80–100: Highly liquid
- 50–80: Moderate liquidity
- <50: Illiquid / specialised asset

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4. Time-to-Liquidate Estimation

Time to Sell = f(Resale Index,
Property Type,
Market Activity Indicators)

Output should be expressed as a **time range**, not a single number.

Input Requirements

Mandatory

- Address or lat-long
- Property type and sub-type
- Size
- Age

Optional

- Rental details
- Legal status indicators
- Property images (bonus)

Sample Output Format

JSON


```
{
"market_value_range": [9500000, 11500000],
"distress_value_range": [7500000, 9000000],
"resale_potential_index": 72,
"estimated_time_to_sell_days": [45, 90],
"confidence_score": 0.68,
"key_drivers": [
"proximity_to_metro",
"standard_2bhk_configuration",
"mid_age_building"
],
"risk_flags": [
"high_micro_market_competition",
"moderate_building_age"
]
}
```

Fraud and Risk Considerations

Common issues to account for:

- Overstated property size
- Incorrect location tagging
- Property type misclassification

Expected safeguards:

- Size sanity checks against locality norms
- Location–property mismatch flags

- Configuration plausibility checks

Evaluation Criteria

Dimension	Weight
Valuation logic	25%
Liquidity modeling	25%
Feature depth	20%
Practical deployability	15%
Explainability	15%

Constraints

- No proprietary transaction datasets assumed
- Allowed:
    - Circle rates
    - Public listings (optional)
    - Reasoned synthetic assumptions

What a Strong Solution Looks Like

Not:

“We trained a regression model to predict price.”

Instead:

“We designed a structured valuation and liquidity framework where location sets the base, property attributes adjust value, market forces define exit risk, and outputs are transparent and range-based.”

Why This Matters

This framework can directly evolve into:

- A core underwriting engine for LAP products
- A real-time valuation API for fintechs
- A credit-focused real estate intelligence layer for India

Final Framing

You are building a **"Bloomberg Terminal for Real Estate Collateral"**
where lenders see **price, risk, and liquidity together**, not in silos.

b) AI-Powered Healthcare Navigator & Cost Estimator for India

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Context: Why This Matters

Across India, particularly in Tier 2 and Tier 3 cities - patients and families face three persistent challenges when seeking medical care:

- Difficulty identifying the *right* hospital or doctor for a specific condition
- Lack of clarity and predictability in treatment pricing
- Wide variation in costs across hospitals, with limited standardization

These issues often lead to delayed medical decisions, financial anxiety, and suboptimal hospital selection.

For lenders and insurers, this lack of transparency creates additional challenges:

- High uncertainty in treatment costs
- Difficulty in confidently pre-approving healthcare loans or coverage

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